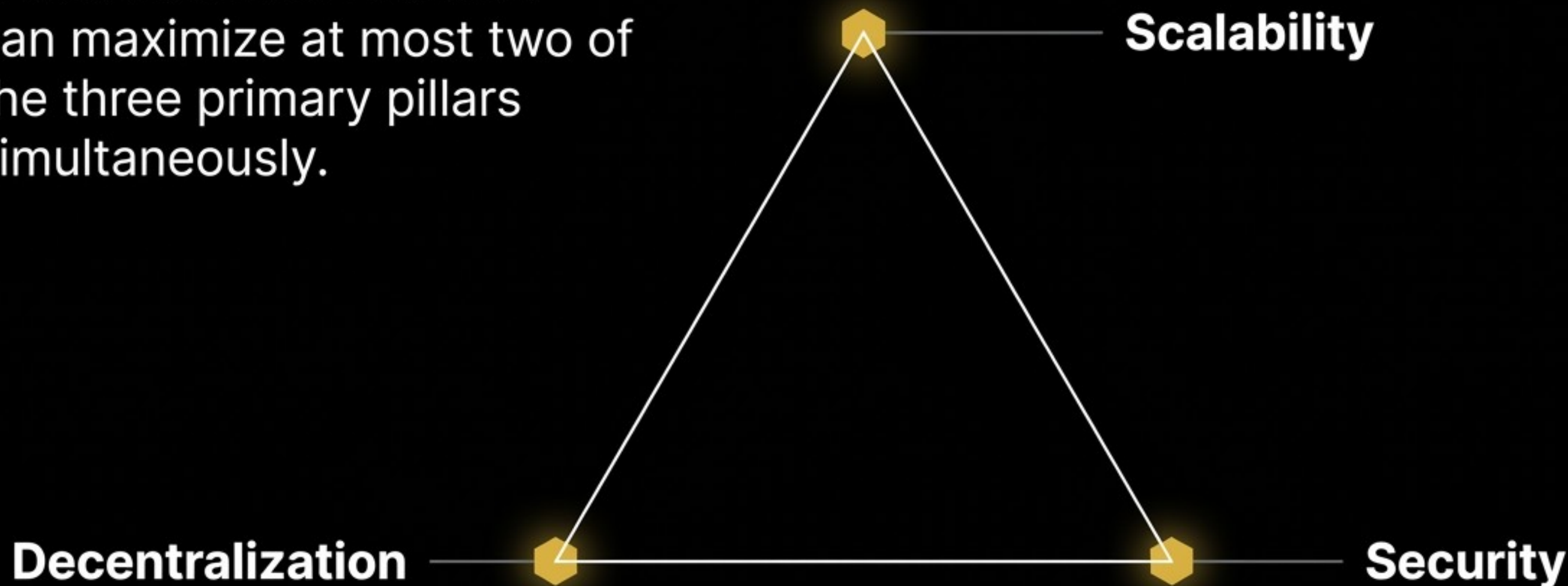


The Blockchain Trilemma

A decentralized network can maximize at most two of the three primary pillars simultaneously.



Formulated by Vitalik Buterin (2017) as an uncompromising law of computing physics for public distributed ledgers.

The Three Pillars Defined

1

Decentralization

The network can be verified and run by everyday users on standard consumer laptops. Avoids reliance on expensive data center oligarchies.

2

Security

Mathematically and economically immune to coordinated Byzantine attacks, 51% reorgs, and validator cartels. Capital cost to attack far exceeds potential profit.

3

Scalability (Throughput)

The system can process thousands of transactions per second (TPS) with sub-second latency and near-zero gas fees.

The Monolithic Spectrum

Decentralization + Security

Bitcoin, Ethereum L1

Trade-off: Low throughput (7–30 TPS), massive gas spikes during network congestion.

Scalability + Security

Solana, Binance Smart Chain

Trade-off: Requires enterprise datacenter hardware (multi-core CPUs, 256GB RAM, fiber connections). Pushes out consumer verification.

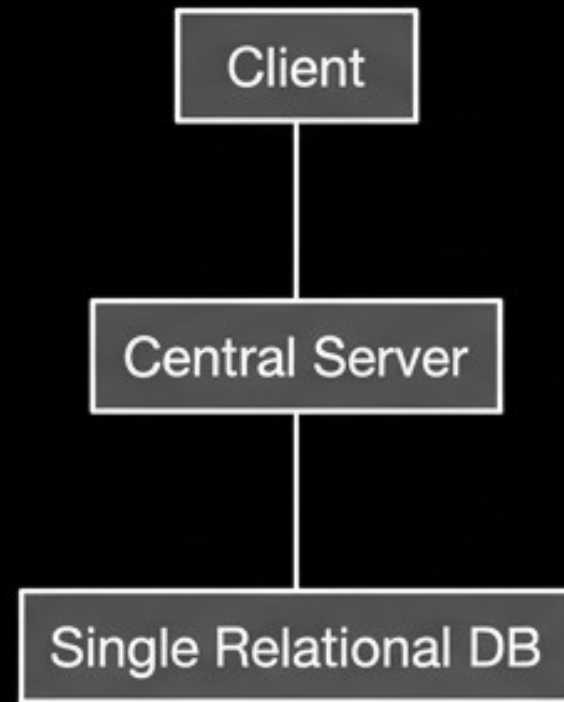
Scalability + Decentralization

Multi-chain Sharding

Trade-off: Fragments security. Individual isolated shards become highly vulnerable to minority validator hijacks.

The Throughput Mystery

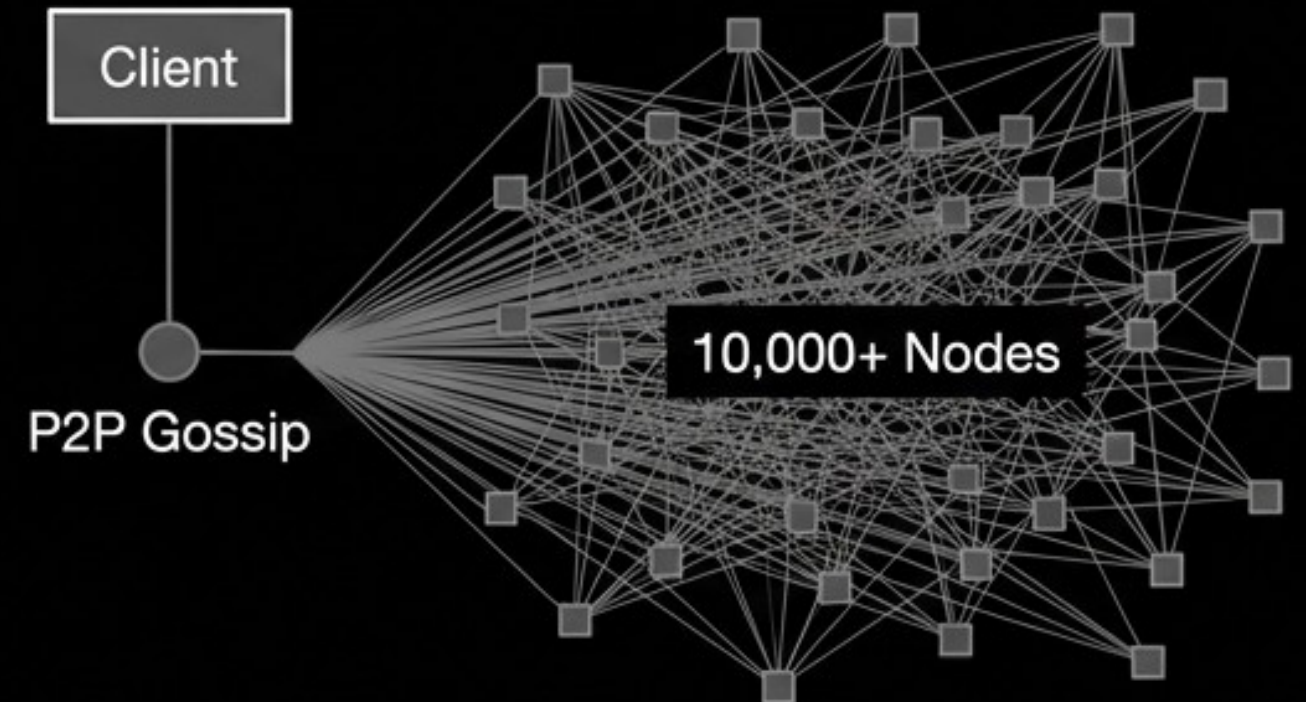
Visa Architecture



24,000 TPS

Zero Byzantine consensus.
Fast, single execution.

Blockchain Architecture

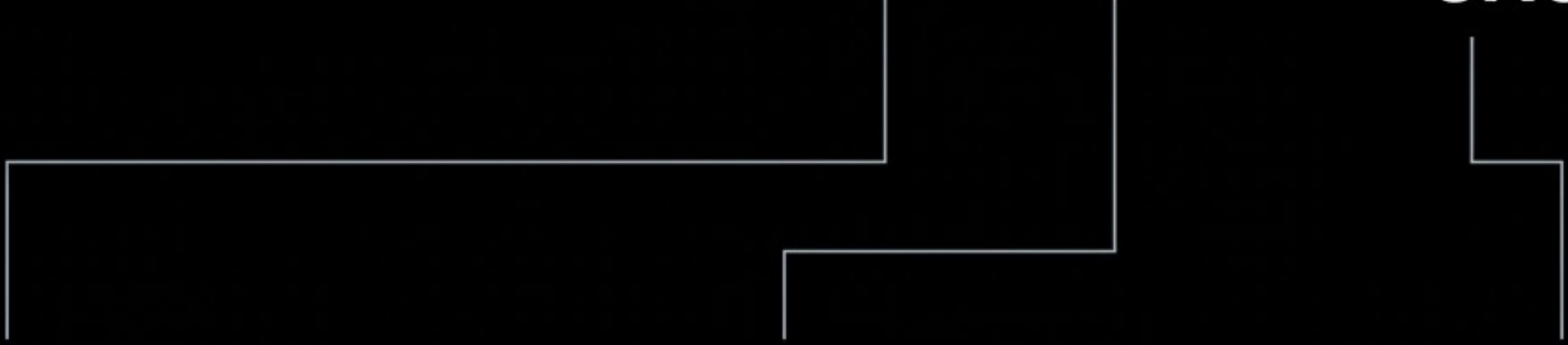


7–30 TPS

Redundant Execution

A swap initiated by Bob in Tokyo is computed identically and redundantly by independent nodes in Berlin, New York, and Jakarta.

The Physics of Network Throughput

$$\text{Throughput} \propto S / (\Delta + T_{\text{exec}})$$


The diagram illustrates the components of the throughput formula. A horizontal line connects the 'S' (Block Size) and the denominator '(\Delta + T_{\text{exec}})'. From the 'S', a vertical line goes down to its definition. From the denominator, a vertical line goes down to a horizontal line that then branches into two vertical lines, each leading to the definition of '\Delta' and 'T_{\text{exec}}' respectively.

S (Block Size)

Pushing more data per block increases transmission time over the physical network.

Δ (Propagation Time)

Latency across submarine fiber-optic cables. If Δ is too high relative to block time, consensus fractures, causing severe spikes in orphan and uncle block rates.

T_{exec} (Execution Time)

The CPU time required to verify cryptographic signatures and execute smart contract bytecode.

Hardware Validator Bottlenecks

Network Bandwidth

Constantly
broadcasting
hundreds of
megabytes
triggers
synchronization
failures on residential
internet.

CPU Execution

Intensive processing
required to evaluate
complex opcodes
complex opcodes
and verify ECDSA
cryptographic
signatures.

Disk I/O & State Bloat

**The ultimate fatal
bottleneck.**

Every transaction
requires
reading/writing the
Merkle Patricia Trie to
local storage
(LevelDB/Pebble).

Disk IOPS is the primary
physical limit choking
monolithic throughput.

The Monolithic Centralization Trap

Naive Scaling

Arbitrarily increasing block sizes to 500MB and slashing block times to 1 second.

State Explosion

The global account state bloats by tens of gigabytes daily, reaching terabytes rapidly.

Home Nodes Crash

Consumer laptops permanently fail to sync with the chain tip.

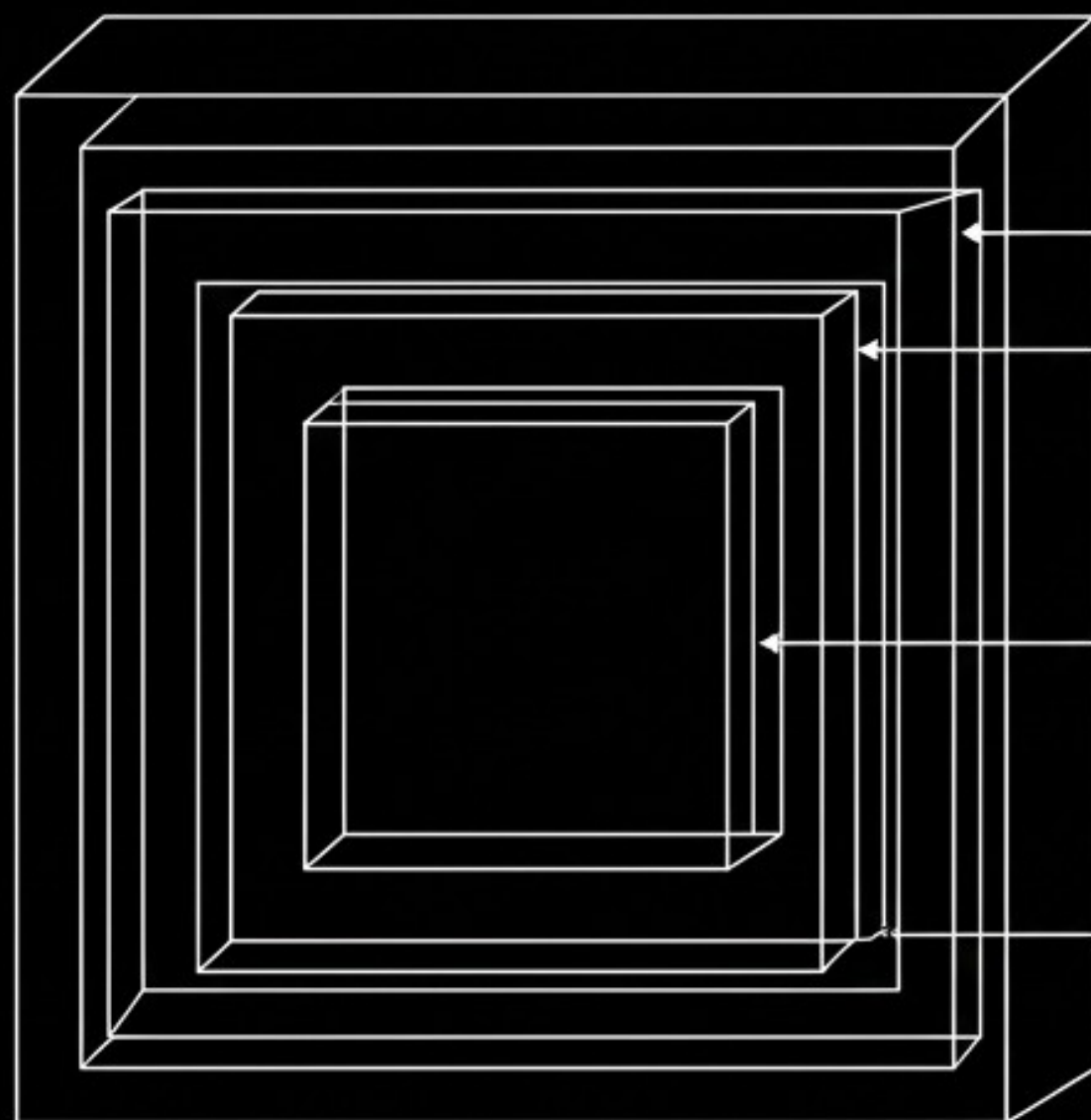
Datacenter Oligarchy

Only elite institutions can afford multi-processor enterprise servers and NVMe storage clusters.

Loss of Sovereignty

The network devolves into a centralized clone, entirely vulnerable to censorship.

Anatomy of a Monolithic Chain (2009–2020)



Execution:

Processing smart contract logic and mutating account balances.

Settlement:

Finalizing transactions, resolving fraud disputes, establishing absolute validity.

Consensus:

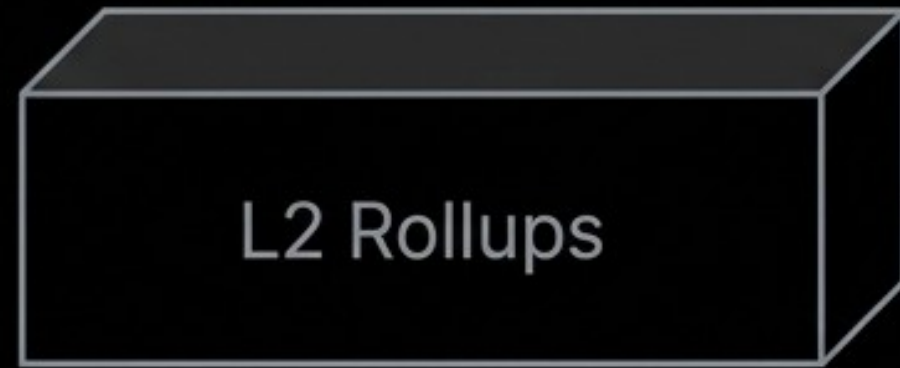
Determining the unalterable chronological order of transactions.

Data Availability (DA):

Guaranteeing all raw transaction data is published and publicly accessible.

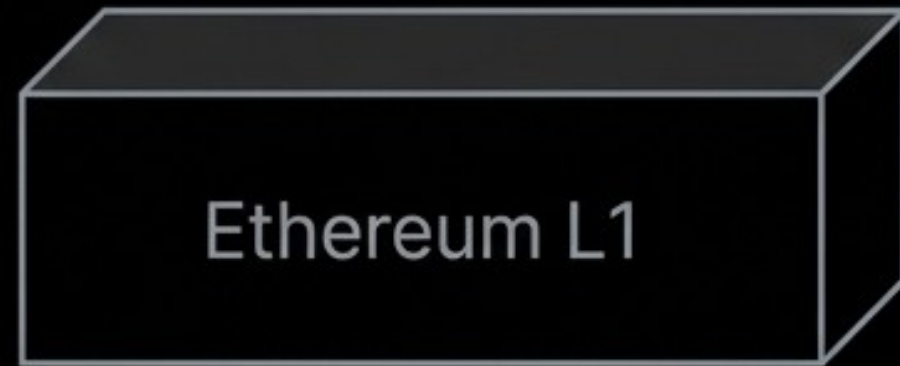
Insight: All four functions fight for the same CPU, RAM, and Disk space on a single node layer, resulting in gridlock.

The Modular Paradigm Shift



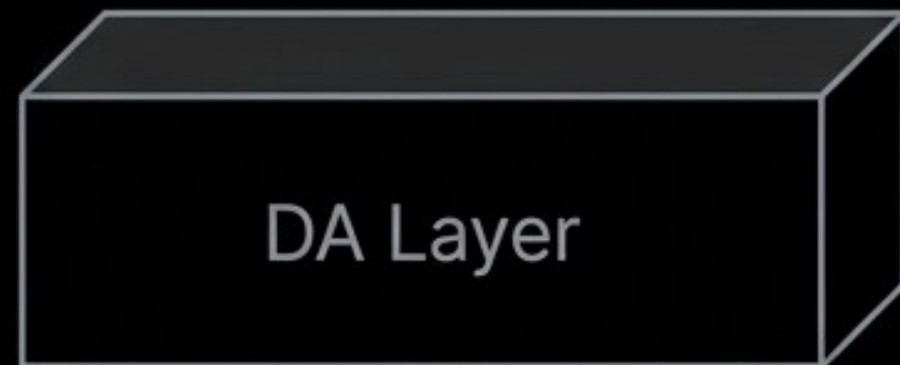
Top Layer: Execution

Layer 2 Rollups (Arbitrum, Optimism).
Highly optimized for off-chain, high-speed transaction processing.



Middle Layer: Settlement & Consensus

Freed from mass computation, L1 acts as the supreme court: ordering transactions, verifying cryptographic proofs, and securing economic finality.



Base Layer: Data Availability

EIP-4844 / Celestia / EigenDA.
Dedicated exclusively to guaranteeing raw transaction data availability without choking L1 execution.

Resolving the Trilemma

Synthesis



- **Decentralization + Security (L1)**

L1 only validates compressed proofs, keeping hardware requirements light. Users can still run nodes at home.

- **Scalability (L2)**

Thousands of TPS executed off-chain. L2 unlocks speed via “Inherited Security” — borrowing L1’s multi-billion-dollar economic shield.

The Bridge



The New Architecture Challenge

If execution happens off-chain, how does L1 trustlessly verify the math without doing the work itself?

NEXT MODULE:
Layer 2 Fundamentals

State Channels, Plasma, Sidechains vs. True Rollups, and the mathematics of Unilateral Exit.